

STAT 231 Midterm 2 Information

Please read the Midterm 1 Information File to remind yourself of the policies and procedures for our exams. This document *primarily* covers differences between the first and second midterm. If you have a question about Midterm 2, please double-check that this has not already been answered in the context of Midterm 1. For the avoidance of doubt, note that the timing and structure of the exam will be the same: 90 minutes with a total of 60 marks spread across Parts A (25%), B (25%), and C (50%). This exam will also be open book.

Coverage

Midterm 2 covers the material in Chapters 1-5 of the Course Notes, with the exception of Sections 2.4 and 5.4. It also covers any additional material covered in Tutorials 1-4, the Introductory R Tutorial, R Tutorials 1-3, and Assignments 1-3. There is a small number of specific topics (such as one-sided confidence intervals and one-sided hypothesis tests) that are mentioned in the Course Notes, but are not covered in STAT 231. This document does not contain a comprehensive list of these excluded topics, as all topics of this kind have been highlighted in class.

The exam will primarily focus on material that was not covered on Midterm 1. However, you are expected to be familiar with all of the above material, and will encounter some questions that rely on knowledge of the earlier chapters. Please do not ask what percentage (or even an approximate percentage) of the exam will cover which chapters; due to the nature of the course material (which is constantly building on itself) we cannot provide an answer to such questions.

Important: You are strongly advised to bring the assignment question files (and your solutions) to the midterm. Questions may directly reference the analyses in the assignment, and so it is assumed you are familiar with the assignments and will have the questions available to reference in your materials.

Material Summary

To aid you in preparing for the exam, here is a list of the ideas and definitions you should know:

Empirical Studies (Sections 1.1-1.2)

- Units, populations and processes.
- Variate and types of variates.
- Response versus explanatory variates.
- Attributes of a population or process.
- Types of studies (sample surveys, observational studies, experimental studies).
- Measures of location: sample mean, median, mode.
- Measures of variability: sample variance, sample standard deviation, range, IQR.
- Measures of shape: skewness, kurtosis.
- Sample percentiles and quantiles. Lower or first quartile, upper or third quartile. Five number summary.
- Numerical summaries for bivariate data: sample correlation, relative risk.

Graphical Summaries (Section 1.3)

- Relative frequency histograms.
- Empirical cumulative distribution function.
- Scatterplots, run charts, and boxplots.

Data Analysis and Statistical Models (Sections 1.4-1.5, 2.1)

- Descriptive statistics.
- Discrete statistical models: $\text{Binomial}(n; \theta)$, $\text{NegativeBinomial}(k; \theta)$, $\text{Poisson}(\theta)$, $\text{Geometric}(\theta)$.
- Continuous statistical models: $\text{Exponential}(\theta)$, $G(\mu, \sigma)$.

Estimation (Sections 2.2, 2.3, 2.5)

- Estimate of a parameter.
- Likelihood functions for discrete and continuous distributions.
- Maximum likelihood estimate (Definition 12).
- Relative likelihood function (Definition 13) and log likelihood function (Definition 14).
- Likelihood function for a random sample.
- Invariance property of maximum likelihood estimates (Theorem 16).

Check the Fit of a Model (Section 2.6)

You should know how to check the fit of a model by:

- Calculating expected frequencies using the assumed model (discrete distributions **only**) and comparing them with observed relative frequencies.
- Examining a graph of the probability density function of the assumed model superimposed on a relative frequency histogram for the observed data (continuous distributions).
- Examining a graph of the cumulative distribution function of the assumed model superimposed on a graph of the empirical cumulative distribution function for the observed data (continuous distributions).
- Examining a Q-Q plot. Note that you will not be asked to construct a Q-Q plot, but you should be familiar with how it is constructed in order to answer questions about what the Q-Q plot tells you about a dataset.

PPDAC (Chapter 3)

You should be familiar with the terminology introduced in this chapter and, in particular, be able to read a study description and answer related questions.

- Types of problems (descriptive, causative, predictive).
- Target population, study population, and sample/sampling protocol.
- Study error, sample error, measurement error.

Estimation (Sections 4.1-4.2)

You should know and/or understand the following:

- Definition of a point estimate of a parameter.
- Definition of a point estimator of a parameter.
- Sampling distribution of an estimator.
- In particular: if we draw a sample of size n from a population Y where $E[Y] = \mu$ and $sd(Y) = \sigma$, what affects the variance of our sampling distribution for $\bar{Y}$?

- **Important:** it is assumed you are familiar with the central limit theorem and associated results from STAT 230. Note that this includes standardizing Gaussian variables. In particular, you should be able to use the fact that if $Y \sim G(\mu, \sigma)$ then $\frac{Y-\mu}{\sigma} \sim G(0, 1)$ to answer questions concerning probabilities associated with Y . You **do not** need to be able to use probability distribution tables for this test, however!

Likelihood Intervals (Section 4.3)

You should be familiar with the following concepts:

- Relative likelihood function.
- 100p% likelihood interval.
- Log relative likelihood function.
- How to obtain likelihood intervals from a graph of the relative likelihood function or the log relative likelihood function.
- Behaviour of likelihood functions, log relative likelihood functions, and likelihood intervals as the sample size increases.
- Guidelines for interpretation of likelihood intervals.
- In particular, you should know how to construct the relative likelihood functions for each of the following: Binomial(n, θ); Geometric(θ); Negative Binomial(k, θ); Poisson(θ); Exponential(θ); $G(\mu, \sigma)$, σ known; $G(\mu, \sigma)$, μ known. You could also be asked to derive the relative likelihood function for a ‘new’ distribution, given its probability function or probability density function. You will not be asked to draw relative likelihood function plots in the exam.

Confidence Intervals (Section 4.4)

Confidence intervals: like likelihood intervals, but different. Make sure you know all the following:

- Definition of a confidence interval.
- Interpretation of a 100p% confidence interval.
- Behaviour of the width of a confidence interval as the sample size, confidence level and (if applicable) population variance change.
- Definition of a pivotal quantity and the pivotal quantities and approximate pivotal quantities detailed in Tables 4.8.1 and 4.8.2 of the Course Notes.
- Use of change of variables to demonstrate a random variable is a pivotal quantity (see example in Tutorial 3).
- How to find a confidence interval given a pivotal quantity or approximate pivotal quantity.
- 100p% confidence interval for the mean of a Gaussian distribution with known standard deviation: $\bar{y} \pm a \frac{\sigma}{\sqrt{n}}$ where $P(Z \leq a) = \frac{1+p}{2}$, $Z \sim G(0, 1)$.
- Approximate 100p% confidence interval for Binomial(n, θ) data: $\hat{\theta} \pm a \sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}}$ where $\hat{\theta} = \frac{y}{n}$ and $P(Z \leq a) = \frac{1+p}{2}$, $Z \sim G(0, 1)$.
- For a Binomial experiment choose the sample size n such that $n \geq \left[\frac{a(0.5)}{\ell} \right]^2$ where $P(Z \leq a) = \frac{1+p}{2}$, $Z \sim G(0, 1)$ to ensure an approximate 100p% confidence interval for θ has width $\leq 2\ell$.
- Approximate 100p% confidence interval for θ for Poisson(θ) data: $\bar{y} \pm a \sqrt{\frac{\bar{y}}{n}}$ where $P(Z \leq a) = \frac{1+p}{2}$, $Z \sim G(0, 1)$.
- Approximate 100p% confidence interval for θ for Exponential(θ) data: $\bar{y} \pm a \frac{\bar{y}}{\sqrt{n}}$ where $P(Z \leq a) = \frac{1+p}{2}$, $Z \sim G(0, 1)$.

Chi-squared and t Distributions (Section 4.5)

You should be familiar with the properties of both distributions:

- Basic properties such as the mean and variance of chi-squared distributed random variables, the mean of t -distributed random variables, and the shapes of their p.d.f.s for various degrees of freedom.
- If $W_1, W_2, \dots, W_n$ are independent random variables with $W_i \sim \chi_{k_i}^2$, then $S = \sum_{i=1}^n W_i \sim \chi_{\sum k_i}^2$.
- If $Z \sim G(0, 1)$, then $Z^2 = W \sim \chi_1^2$, and so if $W \sim \chi_1^2$, then $P(W \geq w) = 2[1 - P(Z \leq \sqrt{w})]$ where $Z \sim G(0, 1)$.
- If $Z_1, Z_2, \dots, Z_n \sim G(0, 1)$ independently, then $S = \sum_{i=1}^n Z_i^2 \sim \chi_n^2$.
- If $W \sim \chi_2^2$, then $W \sim \text{Exponential}(2)$ and so, if $W \sim \chi_2^2$, then $P(W \geq w) = e^{-w/2}$.
- If $Z \sim G(0, 1)$ and $U \sim \chi_k^2$ independently, then $T = \frac{Z}{\sqrt{U/k}} \sim t_k$.
- If T_k follows a t distribution with k degrees of freedom, as k increases the p.d.f. of T_k approaches that of a $G(0, 1)$ random variable.

Likelihood-Based Confidence Intervals (Section 4.6)

This section covered the likelihood ratio statistic, and how it allows us to relate likelihood and confidence intervals.

- Likelihood ratio statistic and its approximate distribution for large n .
- Likelihood-based confidence intervals, such as a 15% likelihood interval is an approximate 95% confidence interval and a 10% likelihood interval is an approximate 97% confidence interval.

Convert a 100 q % confidence interval to a 100 p % likelihood interval:

1. Find c such that $q = P(W \leq c) = P(|Z| \leq \sqrt{c}) = 2P(Z \leq \sqrt{c}) - 1$
2. Then $p = e^{-c/2}$

Convert a 100 p % likelihood interval to a 100 q % confidence interval:

1. $q = 2P(Z \leq \sqrt{-2 \log p}) - 1$

Confidence Intervals for Parameters in the $G(\mu, \sigma)$ Model (Section 4.7)

Be familiar with the following:

- 100 p % confidence interval for the mean μ of a Gaussian distribution with unknown standard deviation σ : $\bar{y} \pm a \frac{s}{\sqrt{n}}$ where $P(T \leq a) = \frac{1+p}{2}$, $T \sim t_{n-1}$ and s is the sample standard deviation.
- 100 p % confidence interval for the variance σ^2 of a Gaussian distribution with unknown mean μ : $\left[\frac{(n-1)s^2}{b}, \frac{(n-1)s^2}{a} \right]$ where $P(W \leq a) = \frac{1-p}{2}$, $P(W \leq b) = \frac{1+p}{2}$, $W \sim \chi_{n-1}^2$.
- For a Gaussian experiment with known standard deviation σ , choose the sample size n such that $n \geq \left(\frac{a\sigma}{\ell} \right)^2$ where $P(Z \leq a) = \frac{1+p}{2}$, $Z \sim G(0, 1)$ to ensure a 100 p % confidence interval has width $\leq 2\ell$.
- Understand the differences between a confidence interval for μ where σ is and isn't known. In particular, that it will be wider when we do not assume σ is known.

Hypothesis Testing (Introduction; Section 5.1)

You should understand the fundamentals of hypothesis testing, including:

- Null hypothesis, alternative hypothesis, test statistic or discrepancy measure.
- p -value and interpretation, including guidelines used in STAT 231 for interpreting p -values (see Table 5.1.1).
- Hypothesis testing for Binomial(n, θ) data, including exact (using the discrepancy measure $D = |Y - n\theta_0|$) and a normal approximation via the Central Limit Theorem.

Hypothesis Testing for Parameters in the $G(\mu, \sigma)$ Model (Section 5.2)

You should be familiar with several specific results relating to the $G(\mu, \sigma)$ model:

- Hypothesis test for $H_0 : \mu = \mu_0$ for the $G(\mu, \sigma)$ model, with unknown standard deviation σ :

$$p\text{-value} = 2 \left[1 - P \left(T \leq \frac{|\bar{y} - \mu_0|}{s/\sqrt{n}} \right) \right]$$

where $T \sim t_{n-1}$ and s = the sample standard deviation.

- Hypothesis test for $H_0 : \sigma^2 = \sigma_0^2$ for the $G(\mu, \sigma)$ model:
 1. Let $u = \frac{(n-1)s^2}{\sigma_0^2}$. Let $U \sim \chi_{n-1}^2$.
 2. If u is large (that is, if $P(U \leq u) > 0.5$) then $p\text{-value} = 2P(U \geq u)$.
 3. If u is small (that is, if $P(U \leq u) < 0.5$) then $p\text{-value} = 2P(U \leq u)$.
- Relationship between confidence intervals and tests of hypothesis. For example, if testing $H_0 : \theta = \theta_0$ results in a $p\text{-value} \leq 0.05$, then a 95% confidence interval for θ will *not* contain θ_0 , and vice-versa.

Likelihood Ratio Tests (Section 5.3)

You should be able to carry out likelihood ratio tests, in particular:

- How the likelihood ratio test relates to the relative likelihood function.
- How to use the likelihood ratio test statistic to test $H_0 : \theta = \theta_0$ for a random variable parameterized by θ .

R Commands

You should be familiar with all R commands introduced in class, in the Introductory R Tutorial, and in R Tutorials 1-3, except for material that is explicitly excluded as discussed in those tutorials (such as functions used for missing data). You may be given R commands and the resulting output (numerical and/or graphical) and be asked to interpret it. You may be asked what the output would be from running a particular R command on a sample of data. You may be asked which of a list of R commands may be appropriate for a particular analysis. You may be asked to identify errors in given R code.